

Forecasting Stock Prices with Time Series Transformers: A Comparative Study of Traditional and Deep Learning Models

Sirong (Rofia) Chen¹

¹rofiaa@umich.edu, Department of Statistics, University of Michigan

April 3, 2025

1 Overview

Forecasting stock prices is a classic yet challenging problem in time series analysis and financial modeling. Traditional methods such as **Simple Moving Averages (SMA)** and statistical models like **ARIMA** have been widely applied due to their simplicity and interpretability. In recent years, neural networks, particularly deep learning models like **Multi-Layer Perceptrons (MLPs)** and **Transformers**, have shown promise in capturing non-linear dynamics and complex temporal dependencies in financial data.

The motivation behind this project is to investigate whether modern time series forecasting architectures, such as Hugging Face's **TimeSeriesTransformerForPrediction**, can offer improved predictive performance over traditional approaches when applied to real-world stock market data. While preliminary results have been obtained using SMA and a simple MLP baseline, the ultimate goal is to explore the benefits of transformer-based probabilistic forecasting.

For this study, I utilize historical daily OHLC (Open, High, Low, Close) data from **Apple Inc. (AAPL)**, focusing on forecasting the next-day closing price. The model inputs include lagged values of OHLC features, while the output is a univariate time series of closing prices. Time-based features (e.g., day of the month, month of the year) may be incorporated to enrich the temporal context.

The key insights expected from this project include:

- Benchmarking the performance of traditional methods (e.g., SMA) and simple neural networks (e.g., MLP) on financial forecasting tasks.
- Assessing the value added by incorporating multiple features (OHLC) in a univariate prediction task.
- Evaluating the potential of transformer-based models for capturing temporal dependencies in noisy, non-stationary stock price data.
- Measuring the quality of forecasts through metrics such as MAE, RMSE, and directional accuracy.
- Visualizing the predicted versus actual stock prices to intuitively assess model alignment and forecasting behavior.

Through this project, I aim to bridge practical forecasting challenges in finance with emerging machine learning tools, ultimately assessing their utility for real-world stock prediction.

2 Prior Work

The field of stock market forecasting has evolved significantly, with advancements spanning traditional statistical methods to contemporary deep learning and transformer-based architectures. These methodologies aim to capture the intricate, nonlinear dynamics inherent in financial markets, thereby enhancing predictive accuracy.

2.1 Literature Review

Hybrid Models: Traditional approaches have combined statistical methods with neural networks to leverage both linear and nonlinear patterns in stock price data. For instance, Khashei and Hajirahimi evaluated hybrid models integrating Auto-Regressive Integrated Moving Average (ARIMA) with Multi-Layer Perceptron (MLP), demonstrating their effectiveness in improving prediction reliability (Khashei and Hajirahimi, 2018).

Deep Learning Techniques: The application of deep learning models has shown promise in capturing temporal dependencies. Mukherjee et al. proposed an LSTM model enhanced by MLP’s comparative analysis to improve stock market predictions (Mukherjee *et al.*, 2024). Similarly, Tang et al. introduced a hybrid approach combining wavelet transform with multi-input LSTM to predict stock movements, effectively capturing both time-frequency domain features (Tang *et al.*, 2024). Additionally, Billah et al. compared different moving average techniques using LSTM networks, achieving a mean absolute error reduction of 23.4% compared to traditional forecasting methods, with an average prediction accuracy of 89.7% (Billah *et al.*, 2024).

Graph-Based Approaches: Wu et al. utilized structural information of financial time series by constructing price graphs, demonstrating that incorporating topological features can enhance stock prediction performance (Wu *et al.*, 2022).

Transformer-Based Models: Recent studies have explored transformer architectures for time series forecasting. Omranpour et al. proposed Higher Order Transformers to enhance stock movement prediction on multimodal time-series data, highlighting the potential of transformers in capturing complex dependencies (Omranpour *et al.*, 2024).

2.2 Potential Methods for Project Goals

To achieve enhanced stock price forecasting, the following methodologies are noteworthy:

- **Transformer-Based Architectures:** Leveraging models such as Hugging Face’s Time-SeriesTransformerForPrediction can improve predictive performance by capturing complex temporal patterns in stock market data (**huggingface*time*series*transformer**).
- **Federated Learning Frameworks:** Kumarappan et al. proposed a federated learning-enhanced model combining MLP for feature extraction and LSTM for sequence modeling. This approach allows collaborative training across institutions without sharing sensitive data, achieving high accuracy (98.3%) and low Root Mean Square Error (RMSE) of 0.0108. However, the implementation details and code base were not publicly provided, limiting reproducibility and practical adoption (Kumarappan *et al.*, 2024).
- **Hybrid Models:** Integrating classical methods like ARIMA with neural networks such as MLP leverages the strengths of both approaches, providing a comprehensive strategy for stock price prediction (Khashei and Hajirahimi, 2018).
- **Graph-Based Models:** Utilizing structural information through price graphs can capture intricate relationships within financial time series data, enhancing predictive capabilities (Wu *et al.*, 2022).

These methodologies underscore the importance of integrating advanced analytical techniques and collaborative frameworks to enhance the accuracy and reliability of stock market predictions.

3 Preliminary Results

3.1 Data Understanding

The **AAPL** dataset includes 6 years of daily trading data of Apple Inc’s stock, capturing both high and low volatility periods. The stock demonstrates moderate daily fluctuations, suggesting the need for models capable of capturing non-stationary patterns and regime changes. Volume outliers may correspond to external market-moving events such as earnings announcements.

Metric	Value
Date Range	2020-04-03 to 2025-04-02
Total Trading Days	1,256
Close Price Range	\$60.35 – \$259.02
Volume Range	23.2M – 374.3M
Average Daily Volatility (Return Std)	0.0181

Table I: Summary statistics for AAPL stock dataset (2020–2025)

Potential Challenges: 1) High noise levels in daily stock prices make accurate forecasting difficult. 2) Volume spikes and price gaps can reduce the reliability of traditional models. 3) Although multiple input features are available (OHLC, volume), the prediction is univariate (future Close).

3.2 Basic Model Evaluation

Model	MAE	RMSE	Directional Accuracy
SMA (10-day)	4.34	5.45	51.7%
MLP	8.72	9.96	54.0%

Table II: Evaluation of baseline models

The **Simple Moving Average (SMA)** serves as a classical statistical baseline. The **MLP** (Multi-Layer Perceptron) slightly outperforms the SMA in directional accuracy, though it exhibits higher absolute error. The MLP better captures trend direction but may overfit due to limited training steps or input structure.

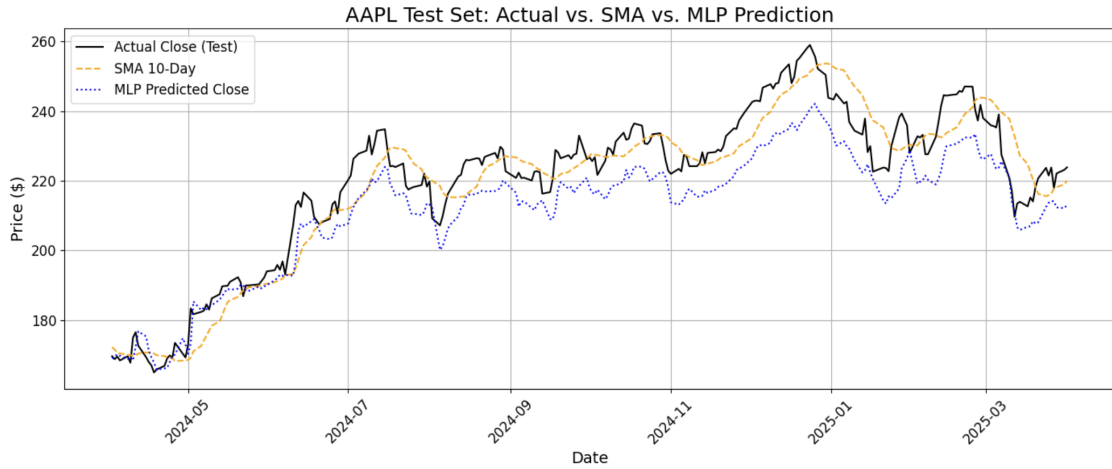


Figure 1: Comparison of actual AAPL closing prices with predictions from the 10-day Simple Moving Average and a Multi-Layer Perceptron on the test dataset (2024–2025).

The line plot illustrates the performance of the MLP model and the 10-day Simple Moving Average (SMA) baseline in forecasting AAPL’s closing prices over the test period. The actual closing prices (black line) show noticeable volatility and trend shifts, particularly in mid-2024 and early 2025. The SMA baseline (orange dashed line) provides a smooth approximation but lags behind rapid price changes. In contrast, the MLP predictions (blue dotted line) respond more dynamically to market movements, capturing some turning points but still underestimating extreme shifts. Overall, both models follow the general trend, with the MLP offering slightly more responsiveness at the cost of increased variance.

3.3 Tools Used and Next Steps

Tools from Class Used So Far:

- Pandas for time-series data processing

- NumPy and Matplotlib for data exploration and visualization
- Scikit-learn for scaling and evaluation metrics
- PyTorch for implementing the MLP model

Tools to Explore Next:

- Hugging Face’s `TimeSeriesTransformerForPrediction` for probabilistic forecasting
- Temporal encoding techniques such as Fourier features
- Model interpretability and feature attribution methods
- Hyperparameter tuning and early stopping to improve generalization

4 Project Deliverables

A successful project will produce a robust and well-documented stock price forecasting pipeline using deep learning models, evaluated against traditional baselines. The final deliverables will include:

- A trained transformer-based time series forecasting model (`TimeSeriesTransformerForPrediction`) for stock price prediction.
- A quantitative evaluation comparing the transformer model with baseline models such as Simple Moving Average (SMA) and a Multi-Layer Perceptron (MLP).
- Visualizations of predictions vs. actual prices on the test set.
- Metrics such as MAE, RMSE, and directional accuracy for assessing forecasting performance.
- A 2-page IEEE-style final report summarizing the methodology, experiments, results, and insights.

Sub-goals include:

- Performing exploratory data analysis and preprocessing of historical OHLC stock data.
- Implementing baseline models (SMA, MLP) and analyzing their effectiveness.
- Integrating time-related features into model inputs (e.g., day of month, month).
- Fine-tuning and evaluating the transformer model using appropriate configurations.
- Assessing forecasting uncertainty and directional trends.

5 Timeline

- **Week 1–2:**
 - Perform exploratory data analysis and preprocessing on the AAPL dataset.
 - Implement and evaluate traditional baselines such as Simple Moving Average (SMA).
 - Build and train a basic Multi-Layer Perceptron (MLP) model.
 - Evaluate and visualize preliminary results using MAE, RMSE, and directional accuracy.
- **Week 3–4:**
 - Integrate the `TimeSeriesTransformerForPrediction` model from Hugging Face.
 - Engineer time features and structure the dataset accordingly.
 - Train and tune the transformer model on the preprocessed data.
 - Compare model performance against MLP and SMA baselines.
- **Week 5:**
 - Refine model and visualizations for final report.
 - Analyze insights from all models and prepare the project summary.
 - Finalize LaTeX report and prepare code/documentation for submission.

References

- Billah, Md Masum *et al.*, (2024). “Stock Price Prediction: Comparison of Different Moving Average Techniques Using Deep Learning Model”, *Neural Computing and Applications*, Vol. 36, pp. 5861–5871. DOI: <https://doi.org/10.1007/s00521-023-09369-0>.
- Khashei, Mehdi and Hajirahimi, Zahra (2018). “A Comparative Study of Series ARIMA/MLP Hybrid Models for Stock Price Forecasting”, *Communications in Statistics - Simulation and Computation*, Vol. 47 No. 8, pp. 2252–2271. DOI: <https://doi.org/10.1080/03610918.2018.1458138>.
- Kumarappan, Jayaraman *et al.*, (2024). “Federated Learning Enhanced MLP–LSTM Modeling in an Integrated Deep Learning Pipeline for Stock Market Prediction”, *International Journal of Computational Intelligence Systems*, Vol. 17, p. 267. DOI: <https://doi.org/10.1007/s44196-024-00680-9>.
- Mukherjee, S. *et al.*, (2024). “Improving Stock Market Predictions Using LSTM and Comparative Analysis”, *AIP Conference Proceedings*. Vol. 3072. No. 1, p. 020023. DOI: <https://doi.org/10.1063/5.0199407>.
- Omranpour, Soroush, Rabusseau, Guillaume, and Rabbany, Reihaneh (2024). “Higher Order Transformers: Enhancing Stock Movement Prediction On Multimodal Time-Series Data”, *arXiv preprint arXiv:2412.10540*, **available at**: <https://doi.org/10.48550/arXiv.2412.10540>.
- Tang, B. *et al.*, (2024). “Stock Movement Prediction: A Multi-Input LSTM Approach”, *Journal of Forecasting*, Vol. 43 No. 5, pp. 1199–1211. DOI: <https://doi.org/10.1002/for.3071>.
- Wu, Jian *et al.*, (2022). “Price graphs: Utilizing the structural information of financial time series for stock prediction”, *Information Sciences*, Vol. 588, pp. 405–424. DOI: <https://doi.org/10.1016/j.ins.2021.12.089>.